

emulators. Test engineers can attach all the devices or start up all the emulators at once, connect to each one, in turn, programmatically, and then run one or more tests. They can also start up the emulator configuration programmatically, run one or more tests, and shut down the emulator.

- **Functional testing:** The MonkeyRunner software can execute an automated start-to-finish test of any Android application. Test engineers can provide the input values with keystrokes or touch events, and view the results as screenshots.
- **Regression testing:** The software can test the app's stability by executing an app and comparing its output screenshots to a set of screenshots that are known to be correct.
- **Extensible automation:** As MonkeyRunner is an API toolkit, test engineers can develop an entire system of Python-based modules and programs for controlling Android devices. Besides using the MonkeyRunner API itself, engineers can use the standard Python OS and sub-process modules to call Android tools such as Android Debug Bridge.

URL: <http://developer.android.com/tools/help/>

monkeyrunner_concepts.html

Sikuli: This is another open source tool that is used

for automating GUI testing. It is an open cross-platform visual platform for creating the scripts, and is oriented towards programming the graphic interface with the help of images or screenshots. Sikuli is a tool that can supplement the testing toolbox. The scripts the test engineer creates with this tool are in the Sikuli Script language, which has a Python (Jython) extension.

URL: <http://www.sikuli.org>

Calabash: Calabash enables test engineers to write and execute automated acceptance tests for mobile apps. It is cross-platform, and supports Android and iOS native apps. Calabash is free, being distributed as open source software, with the company Xamarin backing and developing it.

URL: <http://calaba.sh> 

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...Continued from page no 64

```
user=>(t/union a '[9 10 11])
[1 2 3 4 5 6 7 8 9 10 11]
```

7. **Sorted:** Collections that follow sorted abstraction are assured that their elements will be sorted. Only maps and sets are available in the sorted variant. The main functions here are as follows.

- **rseq**, which returns the sequence in the reverse order:

```
user=>(vec (range 16))
[0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16]
user=>(rseq (vec (range 16)))
(15 14 13 12 11 10 9 8 7 6 5 4 3 2 1)
```

- **subseq**, which returns a *subseq* that's in a particular order:

```
user=>(subseq (sorted-set 1 2 3 4 5 6 7 8) > 4)
(5 6 7 8)
```

- **rsubseq**, which is the same as *subseq*, but in reverse order:

```
user=>(rsubseq (sorted-set 10 20 30 40 50) < 29)
(20 10)
```

What next?

ClojureCLR: ClojureCLR is a port of Clojure to the .NET CLR. It is not a simple cross-compilation; rather, it is maintained separately and aims at providing the same level of usability for CLR as it provides for JVM. Even though ClojureCLR is not as widely used as Clojure, it's well-tested and provides a lot of

options for anyone who wishes to work on it.

ClojureScript: ClojureScript (github.com/clojure/clojurescript) is mainly targeted at JavaScript (technically, ECMAScript 3), and therefore produces code that can be run in all the latest browsers, as well as in other environments that execute JavaScript such as Node.js, CoffeeScript, etc. There is a big difference between Clojure and ClojureScript. This is not only because there is a major difference between the hosts for JavaScript and the JVM, but also due to the fact that ClojureScript is a source-to-source compilation. An overview of the differences between Clojure and ClojureScript is available at github.com/clojure/clojurescript/wiki/Differences-from-Clojure.

Though Clojure has not reached its zenith of success and popularity yet, it's pretty certain that it will do so in a couple of years. Many of the core companies have started adding Clojure to the pre-requisites candidates need to have when it comes to recruiting. So, we can say with certainty that Clojure is indeed the future. 

References

- [1] clojure.org
- [2] stackoverflow.com/3
- [3] 'Clojure in Action' by Amit Rathore
- [4] 'Clojure Programming' by Chas Emerick, Brian Carper and Christophe Grand

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